

Lock down a new Linux server with **one command**

and know the moment someone weakens it

Fresh server



- 🔑 password login
- 👑 root can log in
- 🌐 every port open
- 👻 no audit trail
- 👀 nobody watching

security score

61



After 1 command



- 🔒 SSH keys only
- 🚫 root blocked
- 🧱 firewall: deny all
- 📄 changes recorded
- 📺 monitored 24/7

security score

77

THE PROBLEM

What happens to a new server 🖱️

1 · Goes online



default settings

password login on, ports open

2 · Bots knock



guessing passwords

within minutes, all day

3 · Hardened by hand



10 servers =

10 slightly different setups

4 · Quiet change

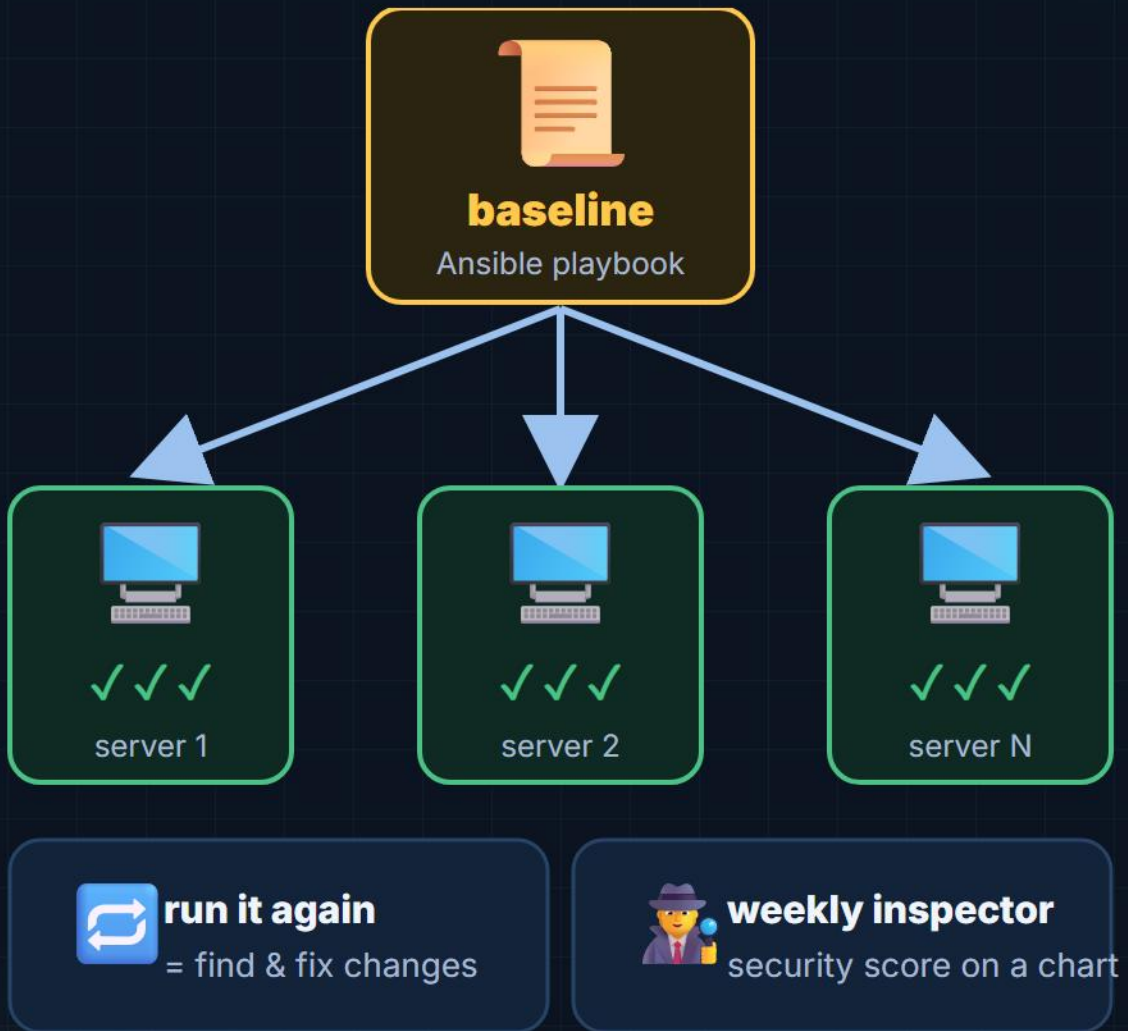


"just for today"

...stays forever. Nobody notices.

THE IDEA

Write the checklist **as code**. Stamp it on every server.



WHEN TO USE IT

Great for... / skip it when...



servers on the internet



many VMs that must be identical



audits ask "prove it is hardened"



long-lived servers that drift over time

Skip it when...



the app runs in containers / Kubernetes pods

(harden the image instead; nothing to log into)



you use a managed service (no server to log into)



a throwaway test VM that lives for an hour

HOW A TEAM USES IT

5 steps, then it runs itself

1



Add the server

one line in the inventory

2



Measure it first

Lynis security score "before"

3



make harden

one server at a time, every change checked first

4



Dashboard watches

alerts if a protection stops or an attacker is banned

5

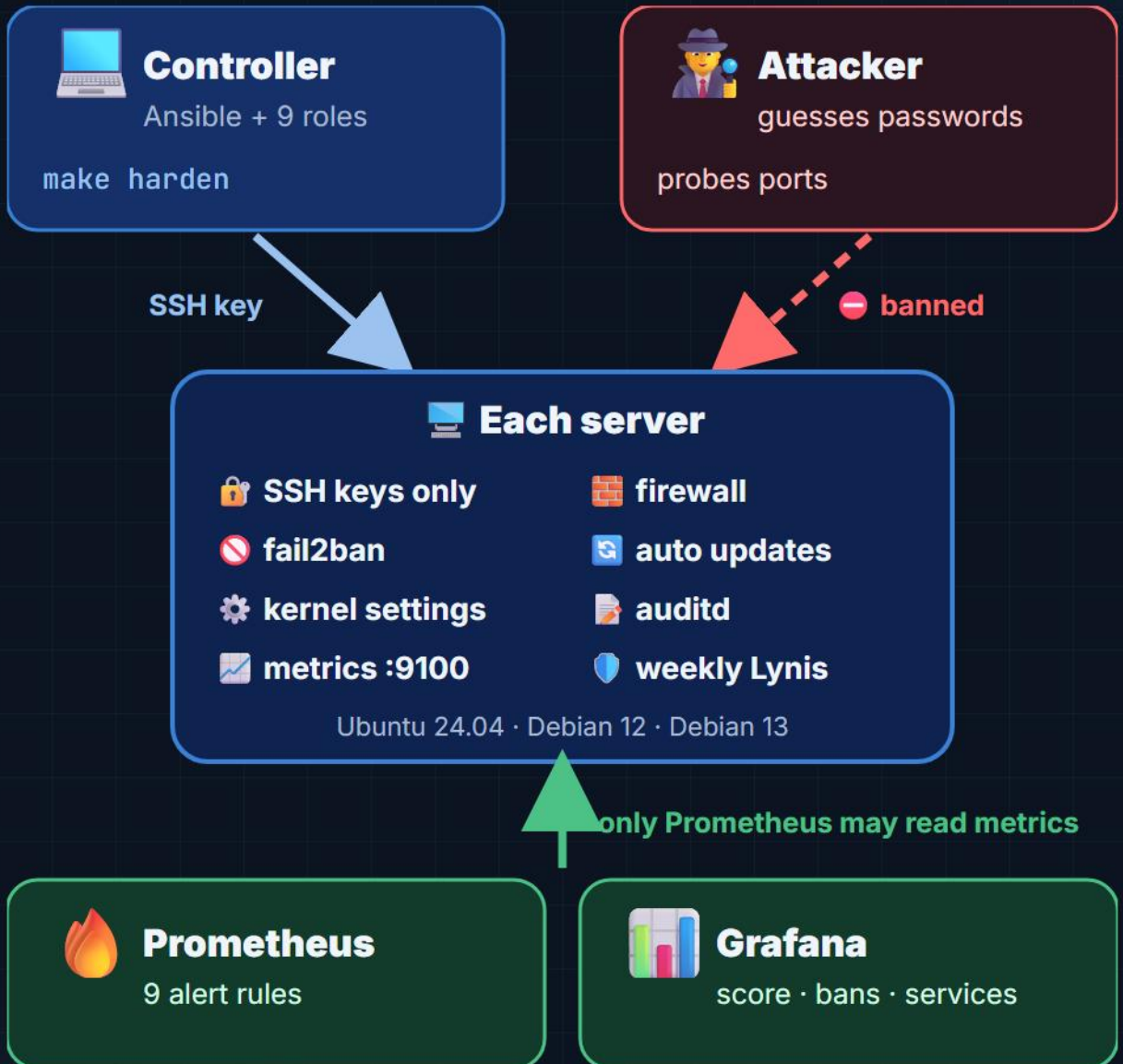


Drift check, any time

lists hand-made changes; re-run puts them back

ARCHITECTURE

How the pieces connect



HOW IT WORKS INSIDE

9 layers, each checked before it lands

1  admins + keys

 stop if no key

2  kernel settings (34)

read → fix only diffs

3  SSH: keys, no root

✓ sshd -t

4  firewall: deny all

✓ nft -c

5  fail2ban

controller never banned

6  security updates

no surprise reboots

7  auditd

who changed what

8  node_exporter

✓ SHA-256 pinned

9  Lynis weekly score

✓ SHA-256 pinned

then: a brand-new SSH login proves you are not locked out

Before → after, on 2 servers

Lynis security score (0–100)

before



61

after



77

same on Ubuntu 24.04 and Debian 12 · 18 fewer suggestions each

password login

~~accepted~~
→ rejected X

root login with a valid key

~~accepted~~
→ rejected X

password-guessing bot

banned in 7 s

alert on the dashboard

after 100 s

2nd run of the playbook

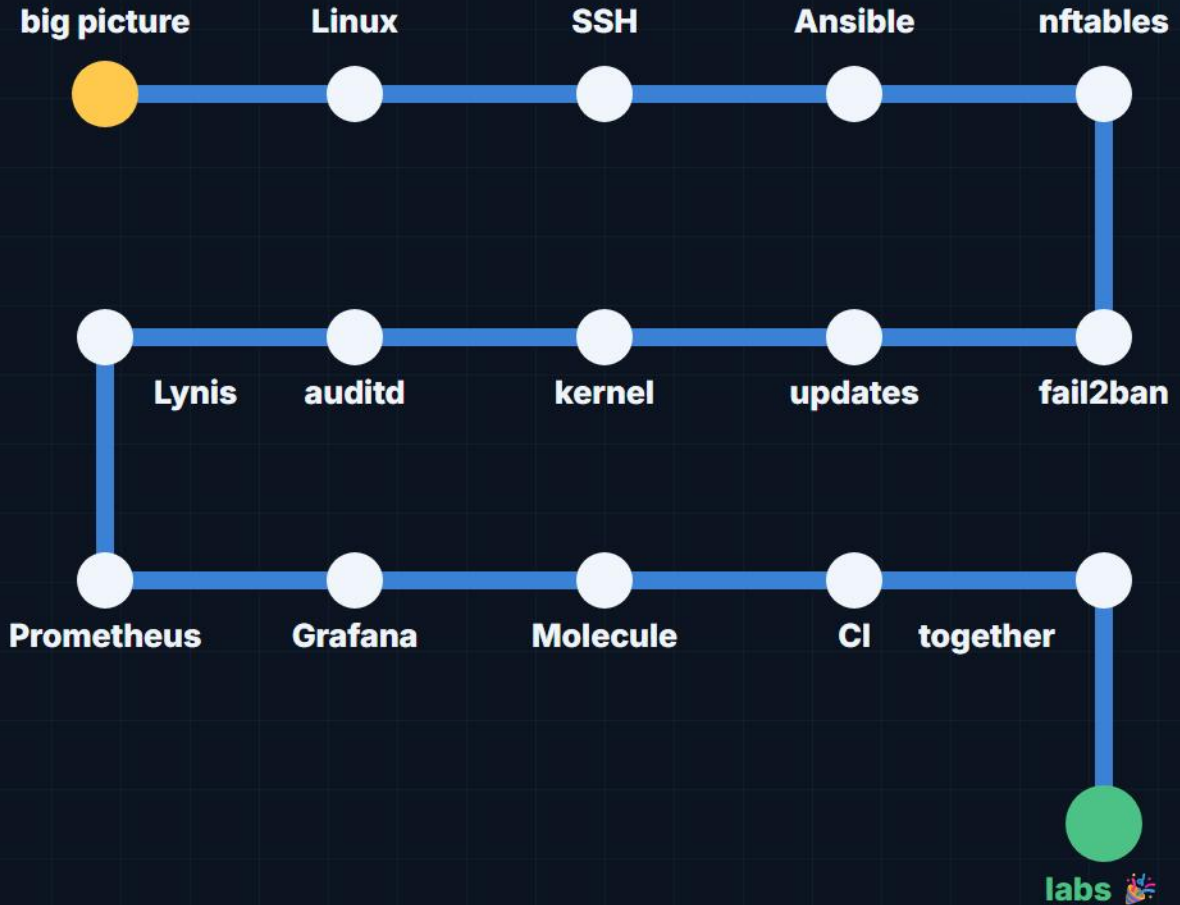
0 changes

3 hand-made changes

found + fixed

FREE STUDY MATERIAL

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page PDF

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hands-on labs

25

interview Q&A

TRY IT YOURSELF

Only Docker needed · ~10 min



\$ make lab-up

2 fresh servers 



\$ make audit LABEL=before

score 



\$ make harden

 the magic



\$ make compare

61 → 77



\$ make drift

 changes?



\$ make e2e

all tests 

YOUR TURN

What's the **first thing** you harden on a new server?



Code + study guide

github.com/sufyanahmadkamboh



All my projects

sufyanahmadkamboh.github.io

Tell me in the comments 🙌

SSH keys? firewall? fail2ban? something else?